true nature of physical reality, we profoundly reconfigure our sense of ourselves and our experience of the universe.

The central concern of this book is to explain some of the most prominent and pivotal of these revisions to our picture of reality, with an intense focus on those that affect our species' long-term project to understand space and time. From Aristotle to Einstein, from the astrolabe to the Hubble Space Telescope, from the pyramids to mountaintop observatories, space and time have framed thinking since thinking began. With the advent of the modern scientific age, their importance has only been heightened. Over the last three centuries, developments in physics have revealed space and time as the most baffling and most compelling concepts, and as those most instrumental in our scientific analysis of the universe. Such developments have also shown that space and time top the list of age-old scientific constructs that are being fantastically revised by cutting-edge research.

To Isaac Newton, space and time simply were—they formed an inert, universal cosmic stage on which the events of the universe played themselves out. To his contemporary and frequent rival Gottfried Wilhelm von Leibniz, "space" and "time" were merely the vocabulary of relations between where objects were and when events took place. Nothing more. But to Albert Einstein, space and time were the raw material underlying reality. Through his theories of relativity, Einstein jolted our thinking about space and time and revealed the principal part they play in the evolution of the universe. Ever since, space and time have been the sparkling jewels of physics. They are at once familiar and mystifying; fully understanding space and time has become physics' most daunting challenge and sought-after prize.

The developments we'll cover in this book interweave the fabric of space and time in various ways. Some ideas will challenge features of space and time so basic that for centuries, if not millennia, they've seemed beyond questioning. Others will seek the link between our theoretical understanding of space and time and the traits we commonly experience. Yet others will raise questions unfathomable within the limited confines of ordinary perceptions.